Evidence Field: Adaptive Covering Programs for Medical Imaging

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Abstract

We introduce Evidence Field, an image-relative encoder and typed spatial-program classifier with reconstructible signed decisions. Learned response fields, structural statistics and parent-conditioned multiplicative scores enter a complete-cover dynamic program. We establish one-fold coverage, finite-grammar optimality, soft-to-hard bounds and signed conservation; class-centred consistency removes common-score offsets. Direct contractions share an adjoint without dense support masks. A histopathology companion obtains AUROC 0.9356/0.9453 and accuracy 84.46%/86.77% at two held-out CAMELYON17 hospitals. Independent readers reconstruct 6,120 class programs with maximum ensemble error below 2.82×10^{-7}. Twenty-four additional evaluated fits test structural, composition and corrected-loss mechanisms under source-only selection. The formulation covers 2D/3D inputs; empirical application evidence here is 2D histopathology.

1 Architectural objective and contribution

Medical acquisition shifts [1, 2] motivate image-relative statistics, typed interactions and exact covering selection. EF replaces CNN-filter [3, 4] or attention-token [5, 6] aggregation, building on radiomics [7], radial/Fourier features [8, 9] and max-affine/multiplicative scores [10, 11].

Relation to spatial and program models. Spatial pyramids pool fixed nested cells [12]; EF selects a class-specific one-fold cover. Sum-product networks compose scoped probability factors [13]; EF maximizes signed complete-region/group scores. Rule ensembles sum signed overlapping predicates [14]; EF constrains terminal supports to an exact cover with parent dependencies. Semiring inference precedes exact grammar reduction [15]. EF couples learned image-relative descriptors, covering selection and independently replayable signed optima; these contracts do not establish superiority over those families.

Measurement contracts address acquisition variation/standardization [16–22]. Lung [23–28], metastasis/nucleus [29–31], molecular/tissue [32–36] and grading studies [37–39] define distinct tasks. Weak-supervision/foundation approaches [40–43] use different training regimes. External evaluation [44, 45] and patient bootstrap [46] motivate source-fixed calibration and uncertainty.

2 Complete model and spatial support contract

Let x occupy a grid $\Omega \subset \mathbb{Z}^d$, $d \in \{2, 3\}$, with $N = |\Omega|$. A declared adapter supplies calibrated measurements and positive axis spacing s_j. The scalar contract uses measurement a and a fixed 3^d local mean $\bar{a}$:

$$q(p) = [a,\ a - \bar{a},\ \partial_z \bar{a},\ \partial_y \bar{a},\ \partial_x \bar{a},\ (\|\nabla \bar{a}\|^2 + 10^{-8})^{1/2}] \in \mathbb{R}^6. \tag{1}$$

Differences are centered internally and one-sided at boundaries; 2D sets the z coordinate/derivative to zero. The optical RGB adapter supplies $q = [o_R, o_G, o_B, |\nabla \bar{o}_R|, |\nabla \bar{o}_G|, |\nabla \bar{o}_B|]$, where $o_c = -\log \mathrm{clip}(x_c, 1/256, 1)$; channel absorption does not assign stain identity [20]. Adapters declare units and provenance.

Independent architecture manuscript.

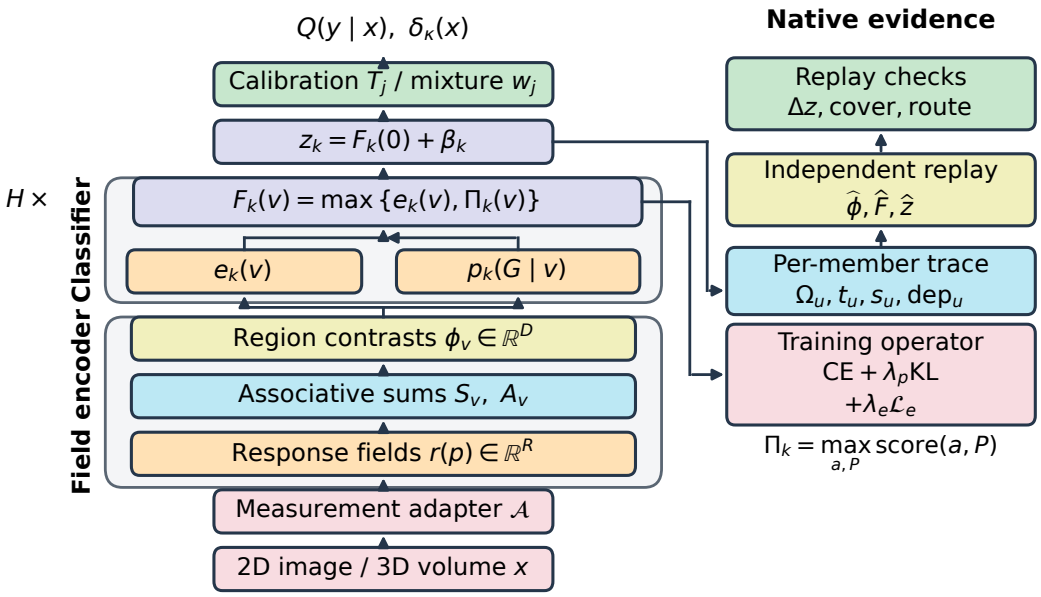


Figure 1: Evidence Field: native measurements, field encoding and depth-H covering classification. Π_k maximizes partition/packing continuations. Original per-member programs support replay; source-fixed calibration forms Q. Training differentiates the smooth recursion.

Occurrence v has support Ω_v, mass $\mu_v = |\Omega_v|/N$, parent $\pi(v)$ and depth $h_v \le H$, $H \ge 1$. Each of A partitions creates $b = 2^d$ disjoint children. The $A = 2$ dictionary registers midpoint and alternating-axis 0.4/0.6 cuts, rounded/clamped inside nondegenerate regions. Occurrences retain parent context; integer rounding may duplicate alternatives. Topology is fixed; scores select partitions/scales with shared descriptor width.

Learned tensor banks. Let $D = 3R(10+2L)+3$. Encoder shapes are $(R, 6)$ for centres/metrics, (R) for mixtures and $(L, 3)$ for frequencies. Own weights have shape (K, M_n, D). Each of $J-1$ interaction-order banks contains affine $(K, M_g, 3D-1)$, member/parent $(K, M_g, D-1)$ and scalar (K, M_g) weights. Including bias,

$$n_\Theta = 13R + 3L + KM_nD + (J-1)KM_g(5D-2) + K. \tag{2}$$

For $R = 8, L = 4, K = M_n = M_g = J = 2$, $D = 435$ and $n_\Theta = 10{,}550$. Buffers and 2D-inactive coordinates are separately recorded.

3 Native image-field encoder

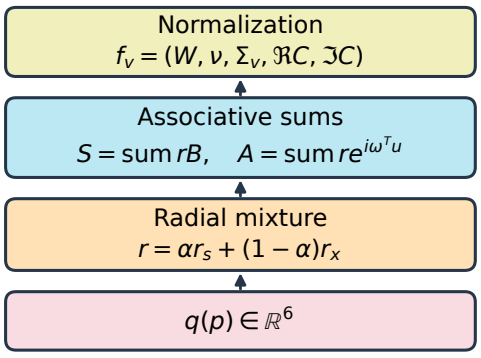


Figure 2: Referenced field encoder.

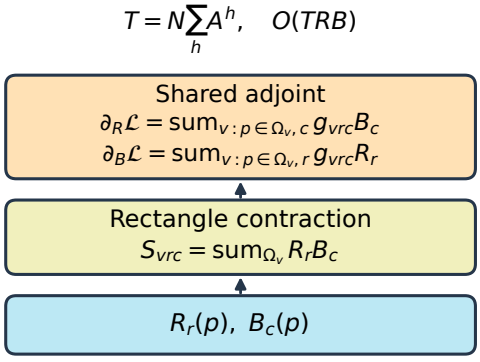


Figure 3: Direct contraction and adjoint.

Absolute and image-relative fields. Source standardizers define $z_s = \mathrm{clip}((q-\mu_s)/\sigma_s, -6, 6)$; current-image moments give $\tilde{q} = (q-\mu_x)/\max(\sigma_x, 0.03)$ and $z_x = \mathrm{clip}((\tilde{q}-\mu_{\mathrm{rel}})/\sigma_{\mathrm{rel}}, -6, 6)$.

For learned centre $c_r \in \mathbb{R}^6$, positive diagonal metric $\Lambda_r = \mathrm{diag}(\mathrm{softplus}\,\rho_r + 10^{-3})$ and mixture $\alpha_r = \mathrm{sigmoid}\,\eta_r$,

$$r(p) = \alpha_r e^{-\frac{1}{2}(z_s - c_r)^T \Lambda_r (z_s - c_r)} + (1 - \alpha_r) e^{-\frac{1}{2}(z_x - c_r)^T \Lambda_r (z_x - c_r)}. \tag{3}$$

The reference is the input distribution. For positive channel gain g and offset t, $\tilde{q}(gq + t) = \tilde{q}(q)$ when the variance floor is inactive. The adapter and absolute branch retain acquisition dependencies [8].

Associative moments and characteristic sums. Normalized global positions $u \in \mathbb{R}^3$ give

$$\begin{aligned} S_{vr} &= \sum_{p \in \Omega_v} r(p)[1, u, \mathrm{vech}(uu^T)] \in \mathbb{R}^{10}, \quad W_{vr} = S_{vr,0}, \\ A_{vrl} &= \sum_{p \in \Omega_v} r(p) e^{i\omega_l^T u(p)}, \quad \omega_l = \omega_{\max} \tanh \kappa_l, \\ C_{vrl} &= e^{-i\omega_l^T m_v} A_{vrl} / \max(W_{vr}, 10^{-6}). \end{aligned} \tag{4}$$

Raw sums add over partitions. For $W_\epsilon = \max(W, 10^{-6})$, centroid $\nu = S_{1:3}/W_\epsilon$, second-moment matrix T from S, and $D_v = \mathrm{diag}(\ell_v)$, define

$$f_{vr} = [W/|\Omega_v|,\ D_v^{-1}(\nu - m_v),\ \mathrm{vech}(D_v^{-1}(T/W_\epsilon - \nu\nu^T)D_v^{-1}),\ \Re C, \Im C]. \tag{5}$$

Zero-mass centroid/covariance entries vanish; virtual 2D z half-width is one. Then $f_v \in \mathbb{R}^{R(10+2L)}$, with midpoint-centred phase and $|C| \le 1$. At fixed response and inactive guard, $\partial C/\partial\omega = i\sum_p r(p)(u(p) - m_v)e^{i\omega^T(u(p)-m_v)}/W$; response gradients include normalization. Finite frequency banks provide spatial sensitivity [9].

Separating moment-equivalent arrangements. On five collinear grid positions $u_j = u_0 + j\Delta$, let $c_j = (-1)^j \binom{4}{j}$. Then $\sum_{j=0}^4 c_j u_j^m = 0$ for $m = 0, 1, 2, 3$, whereas $\sum_{j=0}^4 c_j e^{i\omega u_j} = e^{i\omega u_0}(1 - e^{i\omega\Delta})^4 \neq 0$ when $\omega\Delta \notin 2\pi\mathbb{Z}$. A small ϵc_j perturbation of positive response weights preserves mass, centroid and covariance while changing a sampled signature. This separates response-weighted arrangements; diagnostic value requires real-source ablations.

Support reuse and referenced features. Parent/root contrasts yield

$$\phi_v = \mathrm{clip}([f_v, f_v - f_{\pi(v)}, f_v - f_0, \mu_v, h_v/H, 1], -6, 6), \qquad D = 3R(10 + 2L) + 3. \tag{6}$$

The root references itself. Registry, clipping, guards and coordinate transforms are replayed.

4 Typed cross-scale group classifier

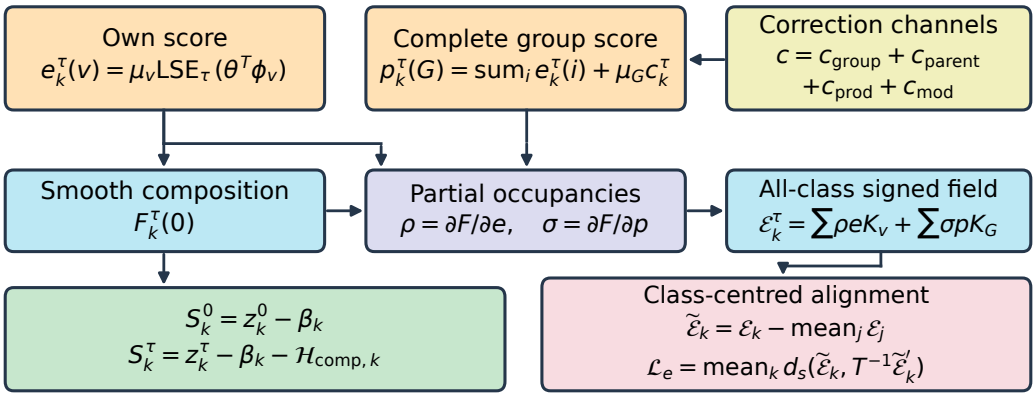


Figure 4: Complete-score operands, partial composition occupancies, signed conservation and class-centred alignment. Group scores retain their member anchors.

Class k has M_n own-region types. With $\bar{\phi}$ denoting the descriptor without its intercept,

$$e_k(v) = \mu_v \max_m \theta_{km}^T \phi_v. \tag{7}$$

A production at v permits connected child groups G of orders two to J. Connectivity uses face adjacency in the partition's child grid. Let $\mu_G = \sum_{i \in G} \mu_i$,

$$\phi_G^\mu = \sum_{i \in G} \mu_i \phi_i / \mu_G, \qquad d_G = \sum_{i \in G} \mu_i |\bar{\phi}_i - (\phi_G^\mu)_{\text{no intercept}}| / \mu_G, \quad \psi_{Gv} = [\phi_G^\mu, d_G, \phi_v].$$

For each order $o = |G|$, M_g types define

$$\begin{aligned} c_{kot}(G \mid v) &= \xi_{kot}^T \psi_{Gv} + \gamma_{kot} \big[1 + \tanh(b_{kot}^T \bar{\phi}_v)\big] \prod_{i \in G} \tanh(a_{kot}^T \bar{\phi}_i), \\ p_k(G \mid v) &= \sum_{i \in G} e_k(i) + \mu_G \max_t c_{kot}(G \mid v). \end{aligned} \tag{8}$$

Shared projections ensure member-permutation invariance. Order-specific banks and parent affine/product terms supply cross-scale interactions [11], bound to declared supports and preserved anchors.

Four signed correction channels. Partition $\xi = [\xi_{\text{group}}, \xi_{\text{parent}}]$ and write $m = \gamma \prod_{i \in G} \tanh(a^T \bar{\phi}_i)$. The selected correction is exactly $c = \xi_{\text{group}}^T [\phi_G^\mu, d_G] + \xi_{\text{parent}}^T \phi_v + m + m \tanh(b^T \bar{\phi}_v)$. Anchors, four corrections and bias replay separately. Parent/root contrasts remain in descriptor ablations.

Interaction property. For fixed type/contrast branch, the product's order-o mixed derivative contains $a^{\otimes o} \mu_G \gamma (1 + \tanh b^T \bar{\phi}_v)$ times member sech^2 factors. Nonzero factors yield interactions absent from independent additive scores.

5 Adaptive covering recursion and exact execution

For candidate partition a at v, let $\mathcal{P}_a(v)$ contain all disjoint permitted group packings. Unmatched children are denoted $U(P)$. Leaves use $F_k(v) = e_k(v)$; internal occurrences use

$$F_k(v) = \max \left\{ e_k(v), \max_{a, P \in \mathcal{P}_a(v)} \left[\sum_{G \in P} p_k(G \mid v) + \sum_{u \in U(P)} F_k(u) \right] \right\}, \qquad z_k = F_k(0) + \beta_k. \tag{9}$$

Groups terminate member subtrees; unmatched children recurse. Packings are registered before learning; class scores select partition, packing, scale and types.

Proposition 1 (Cover and registered optimum)**.** *Every selected program covers Ω exactly once by its score-unit supports. The recursion returns the maximum over this finite grammar.*

Proof. Own supports, disjoint groups and unmatched-child covers partition each selected region. Depth induction gives one-fold coverage and exhaustive maximization. References are dependencies, not extra score supports. □

Proposition 2 (Nested families and rule challenges)**.** *Adding partitions or groups cannot reduce the optimum when common descriptors and scores are fixed. Removing productions cannot increase it. Zero group corrections embed the own-region covering family.*

Proof. The maximized set expands or contracts. With zero corrections, each group equals recursively selectable member anchors. □

For child i in partition a, define

$$B_{ka}(i) = \max\{F_k(i), \max_{G \ni i} p_k(G \mid v)/|G|\}, \qquad U_k(v) = \max_a \sum_{i \in \text{ch}_a(v)} B_{ka}(i). \tag{10}$$

Member B sums bound unmatched/group scores, including negatives. The own choice skips packing reduction if $e_k(v) > U_k(v) + \epsilon_v$, $\epsilon_v = 32 \epsilon_{\text{dtype}} (1 + |e_k(v)| + |U_k(v)|)$, after dense scoring.

6 Differentiable training, observation and uncertainty

Smooth training and hard decisions. Cross-entropy learns all declared banks. Type maxima use $\tau \log \sum \exp(s/\tau)$ before mass scaling; composition uses $\tau \mu_v$. Inference uses hard maxima. Source/local validation selects state and calibration.

Proposition 3 (Soft-to-hard score bound)**.** *For fixed finite descriptors and parameters, depth H, at most A alternatives, at most Q packings per partition and $M_n, M_g \geq 1$,*

$$0 \leq z_k^\tau - z_k^0 \leq \tau\{\log M_n + \log M_g + H\log(1+AQ)\} = \varepsilon_\tau. \tag{11}$$

Proof. Log-sum-exp adds at most temperature times $\log m$. Type gaps scale by covered mass. Group/unmatched masses partition parent mass; induction adds $\tau\mu_v \log(1+AQ)$ per depth and the type gaps once over terminal mass. Root mass is one. □ A hard class margin above ε_τ preserves its winner under this fixed-state smooth reduction.

All-class signed score fields. Let $\mathrm{LSE}_t(s) = t\log\sum_j e^{s_j/t}$ and $\mathrm{LSE}_0(s) = \max_j s_j$. Smoothed own and *complete* group scores e_k^τ, p_k^τ enter $\mathrm{LSE}_{\tau\mu_v}$ composition as separate operands, with occupancies $\rho_{kv}^\tau = \partial F_k^\tau(0)/\partial e_k^\tau(v)$ and $\sigma_{kaG}^\tau = \partial F_k^\tau(0)/\partial p_k^\tau(a, G \mid v)$. Cloned operands retain parameter ancestry; total derivatives through group anchors would double-count allocation.
Let $U_{aG} = \bigcup_{i\in G}\Omega_i$ and define unit-integral kernels $K_v(p) = \mathbf{1}[p \in \Omega_v]/|\Omega_v|$ and $K_{aG}(p) = \mathbf{1}[p \in U_{aG}]/|U_{aG}|$. For every class,

$$\begin{aligned} \mathcal{E}_k^\tau(p) &= \sum_v \rho_{kv}^\tau e_k^\tau(v) K_v(p) + \sum_{a,G} \sigma_{kaG}^\tau p_k^\tau(a,G) K_{aG}(p), \\ S_k^\tau &= \sum_p \mathcal{E}_k^\tau(p). \end{aligned} \tag{12}$$

Hard fields allocate selected expressions; soft fields allocate expected complete scores, excluding bias. Shared references remain dependencies beyond score supports.

Proposition 4 (Signed integral and composition entropy)**.** *Let $r_k^\tau(v)$ be the probability of reaching occurrence v and π_{kv}^τ its local distribution over own termination and all partition/packing choices. Then*

$$z_k^\tau - \beta_k = S_k^\tau + \mathcal{H}_{\mathrm{comp},k}^\tau, \qquad \mathcal{H}_{\mathrm{comp},k}^\tau = \sum_{v:h_v<H} r_k^\tau(v)\tau\mu_v H(\pi_{kv}^\tau) \geq 0. \tag{13}$$

At hard inference, $\sum_p \mathcal{E}_k^0(p) = z_k^0 - \beta_k$.

Proof. $\mathrm{LSE}_t(s) = \sum_j \pi_j s_j + tH(\pi)$ for $\pi = \mathrm{softmax}(s/t)$. Recursively weight local identities by reach probabilities. Unmatched children recurse; groups and own choices terminate, with partial derivative occupancies. Unit-integral kernels allocate their scores. At $t = 0$ the selected covering expression has no entropy residue. □
Type entropy is already contained in smooth unit scores. Only composition entropy is added to their signed integral; both residues vanish under hard reductions. Binary field differences conserve the non-bias class margin.

Class-relative signed stability objective. Let T be an axial quarter turn followed by optional reflection. Recompute the model on Tx and align its class fields by T^{-1}; the inverse reflects first and rotates inversely. Define $\widetilde{\mathcal{E}}_k = \mathcal{E}_k - K^{-1}\sum_j \mathcal{E}_j$ pointwise, retaining the common field separately for score accounting. For equally shaped signed maps A, B, use

$$\begin{aligned} d_{\mathrm{s}}(A,B) &= \frac{\|A-B\|_1}{\|A\|_1 + \|B\|_1 + \epsilon}, \quad \epsilon > 0, \\ \mathcal{L}_e &= \frac{1}{K}\sum_k d_{\mathrm{s}}(\widetilde{\mathcal{E}}_k^\tau(x), T^{-1}\widetilde{\mathcal{E}}_k^\tau(Tx)). \end{aligned} \tag{14}$$

Triangle inequality gives $0 \leq d_{\mathrm{s}} \leq 1$; two zero maps give zero. Centering cancels any aligned class-common field g and commutes with T^{-1}. In particular, a common intercept offset adds C/N to fields and C to logits without changing the corrected loss. Its hard integral is the class-centred non-bias logit; original fields retain Eq. (13). With $P = \mathrm{softmax}(z^\tau/T_p)$ and $T_p > 0$, the batch-averaged objective is

$$\begin{aligned} \mathcal{L} = {} & \tfrac{1}{2}\{\mathrm{CE}(z^\tau(x), y) + \mathrm{CE}(z^\tau(Tx), y)\} \\ & + \frac{\lambda_p}{2}\{\mathrm{KL}(P(x)\|P(Tx)) + \mathrm{KL}(P(Tx)\|P(x))\} \\ & + \lambda_e \mathcal{L}_e. \end{aligned} \tag{15}$$

The implemented signed objective uses $T_p = 1$, $\lambda_p = 0.05$, $\lambda_e = 0.02$ and retains mixed derivatives. Setting both coefficients to zero recovers paired cross-entropy. The primary histopathology

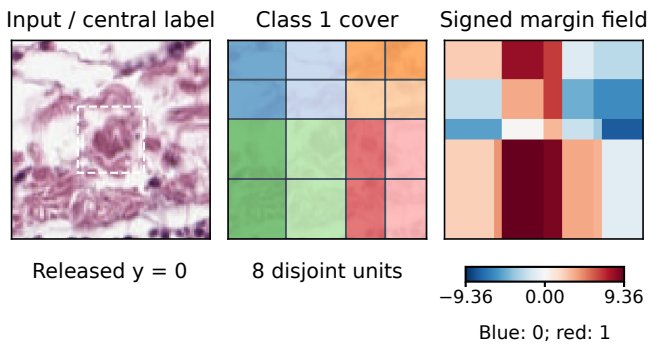
ensemble uses paired cross-entropy. Section 9 reports the separate completed original-signed and class-centred-signed comparisons, preserving these fit identities.

Observation and hospital interfaces. Records bind native inputs/sample counts. Source validation fixes T_j, κ [47–49] for $Q = \sum_j w_j \operatorname{softmax}(z_j/T_j)$, $w_j \geq 0$, $\sum_j w_j = 1$, $T_j > 0$, and review $\delta_\kappa = \mathbf{1}[\max_k Q_k < \kappa]$. Adaptation declares updated banks, disjoint training/validation patients and state/calibration; coverage/error require measurement.

7 Native evidence and independent verification

Independent reconstruction contract. Hard records bind units, supports/types/routes, dependencies, bias and state to $\widehat{z}_k = \beta_k + \sum_{u \in \mathcal{U}_k} s_k(u)$ and $\widehat{P}_k = \operatorname{softmax}(\widehat{z}/T)_k$. A fresh NumPy/SciPy process checks descriptors, scores, grammar, one-fold covers, optima and near ties. Group anchors count once; references are dependencies. Export every ensemble member's state/program, temperature and weight to reconstruct $\widehat{Q} = \sum_j w_j \operatorname{softmax}(\widehat{z}_j/T_j)$; logit averaging does not reproduce it.

Evaluation binding and evidence challenges. Input/state and source-lock identities bind predictions. Grammar challenges preserve descriptors; pixel interventions recompute references. Clinical causality requires assessment.



Signed totals, seed 11

	Class 0	Class 1
Anchors	+2.896	+1.117
Group affine	+3.550	+5.645
Parent affine	+3.046	+3.122
Product	-0.660	+0.144
Modulation	-0.472	+0.124
Bias	+0.005	-0.005
Sum = logit	**8.364**	**10.147**

Raw ensemble P(1) = 0.8876

Figure 5: Real incorrect prediction: dashed central-32^2 label $y = 0$, seed 11's one-fold 96^2 class-1 cover, and bias-free signed field $N(\mathcal{E}_1^0 - \mathcal{E}_0^0)$. Colours identify units, not cells. Full-precision terms give $z_0 = 8.36403762$, $z_1 = 10.14726239$; display values are rounded. All three members give raw $Q_1 = 0.88756339$, an incorrect class-1 decision.

The first panel-order ensemble error is displayed with fixed seed 11. Logit-decomposition errors are $\leq 5.23 \times 10^{-7}$ (seed 11), $\leq 2.02 \times 10^{-6}$ (all members). Replay does not establish diagnostic correctness.

Public reproduction artifact. `https://github.com/isararsiddique/evidence-field` supplies frozen code/reader, three pathology checkpoints, configurations and the example. `download_artifacts.py` verifies payloads; `reproduce_example.py` checks programs/mixture or CPU inference (`-with-model`). The inventory separates example replay, saved statistics and omitted images/full-row manifests.

Algorithm 1: learning and independent deployment. Declare $(\mathcal{A}, s, \mathcal{G}, \Theta)$; encode q, r, S, A, ϕ and typed scores. Compose e^τ, p^τ; derive $\rho, \sigma, \mathcal{E}^\tau$ and optimize Eq. (15). Lock state/calibration, trace F^0 and export signed programs/bias. Independently recompute descriptors, scores, grammar, covers and optima; replay each member/mixture and return z, Q, δ_κ, programs and binding receipts.

8 Implementation and technical validation

Direct rectangular contractions and their adjoint. Direct sums $S_{vrc} = \sum_{p \in \Omega_v} R_r(p) B_c(p)$ use support slices rather than a dense mask. Their shared-array adjoint, for upstream g_{vrc}, is

$$\frac{\partial L}{\partial R_r(p)} = \sum_{v: p \in \Omega_v} \sum_c g_{vrc} B_c(p), \qquad \frac{\partial L}{\partial B_c(p)} = \sum_{v: p \in \Omega_v} \sum_r g_{vrc} R_r(p).$$

Table 1: Hash-bound CAMELYON17 checks: two source loss inputs, both held-out centres for replay. Finite-difference error is relative; others absolute. Loss checks have no optimizer steps.

Check	Cases	Maximum error
Shared-offset loss	48 checks, 2 inputs	1.97×10^{-13}
Loss finite differences	24	1.08×10^{-9}
Three-seed probability	340 images, both centres	2.81×10^{-7}
Native prediction logits	3,060 member-view records	4.16×10^{-6}
Panel descriptors / optima	6,120 class programs	1.19×10^{-7} / 6.38×10^{-6}

The adjoint supports mixed derivatives; FP64 fields/sums yield FP32 descriptors. With $B = 10+2L$ and $T = \sum_v |\Omega_v| = N\sum_{h=0}^{H} A^h$ (each partition covers its parent), arithmetic is $O(TRB)$; $T = 7N$ at $A = 2, H = 2$. Field/sum storage is $O(N(R+B)+VRB)$, excluding measurements, scores and autodiff. Dense contraction needs $O(NVRB)$ arithmetic and $O(NV)$ masks. These arithmetic and storage bounds describe the operator. They do not substitute for end-to-end runtime or energy measurements, which include encoding, replay and training differentiation.

Score and recursion cost. For $V = \sum_{h=0}^{H}(Ab)^h$ occurrences, $P = A\sum_{h<H}(Ab)^h$ productions and G groups per production, candidate arithmetic is $O(VKM_nD + PGKM_gJD)$ and packing arithmetic is $O(PKQ(b+G))$. At $A = 2, H = 2, J = 2$, the 3D grammar has $(V, P, G, Q) = (273, 34, 12, 108)$; larger order banks remain optional. Fields/classes/supports parallelize; recursion has depth dependencies. At $A = 2$, 3D dictionaries grow from $V = 273$ to 4369 and 69905 at depths 2/3/4. The histopathology study uses depth two. The gate skips packing reductions after scoring, not image encoding; direct storage does not remove registry growth.

9 Trained controls and medical application evidence

Source-only training and evaluation. The histopathology companion [50] trains six from-scratch recipes, seeds 11/29/47, for 16 epochs (18 fits). CAMELYON17 [29] centres 0/3/4 supply 271,010 training, 24,982 validation and 40,004 source-test patches; centres 1/2 supply 34,904/85,054 patches from 8/9 development-disjoint patients. All 455,954 patches are included. The binary lymph-node task has two held-out hospitals within one dataset. The wltjr1007 CAMELYON17-WILDS mirror supplies inputs; decoded-image/LFS checks do not authenticate original-publisher bytes. Source training/validation/test contain 20/3/3 patients: retained training IDs, then alternating sorted remaining source IDs, without labels. The artifact records exact IDs and manifest hashes. Previously examined targets make this source-locked replication, not prospective validation or the standard WILDS protocol.

Shared inputs are 96×96 RGB. AdamW uses EF/baseline rates 0.003/0.001, decay 10^{-4}, cosine scheduling, clipping 5, batch size 64 and unweighted CE. Comparators are ResNet18 and a four-block, width-128 ViT with 8×8 patches. Source validation selects state, objective, calibration and review thresholds. Four EF objectives versus one CNN/ViT configuration leave unequal tuning; held-out outcomes do not select models.

Table 2: Histopathology companion: raw three-member ensemble AUROC / accuracy (%), fixed argmax. N counts patches, parentheses patients; EF is source-selected paired CE.

Evaluation	N (patients)	EF	CNN	ViT
CAMELYON centre 1	34,904 (8)	0.9356 / 84.46	0.7352 / 58.94	0.8502 / 76.13
CAMELYON centre 2	85,054 (9)	0.9453 / 86.77	0.7656 / 67.60	0.7924 / 67.27

EF's 95% patient-bootstrap AUROC intervals (2,000 resamples) are $[0.8311, 0.9821]$/$[0.9046, 0.9630]$. Paired EF–CNN/EF–ViT difference intervals are $[0.0972, 0.4443]$/$[0.0051, 0.2523]$ (centre 1), $[0.1078, 0.4319]$/$[0.1137, 0.2993]$ (centre 2). Whole-patient resampling retains patch multiplicity. These exploratory, multiplicity-unadjusted intervals and unequal tuning/small patient counts constrain superiority claims.

Trained mechanism and corrected-loss comparisons. Seven mechanism triplets and three centred-loss fits add 24 evaluated fits (16 epochs each); one learning-rate fit is source-only. Every new row is separately trained: the direct-contraction row is three new fits, not numerical reevaluation of the dense checkpoints. All 72 prediction files and three patient-

statistics reports are complete; external scores do not select a replacement primary model. Gaussian closure derives signatures from moments using the same head/backend. Gaussian closure and the fixed partition both exceed primary EF at both hospitals on point estimates. Learned signatures and adaptive partitions therefore have unestablished predictive benefit here; their mathematical expressivity and covering properties are separate. Direct-full-minus-closure intervals are $[-0.0175, 0.0016]/[-0.0494, 0.0065]$. Parent removal retains contrasts; node-only removes groups; capacity effects are mixed. Centred-minus-original signed changes are $-0.0012/+0.0043$, with intervals $[-0.0064, 0.0021]/[-0.0074, 0.0260]$. Offset invariance does not establish predictive/stability gains.

Table 3: Three-member raw-probability ensemble AUROC; registered n_Θ per member. No-parent masking leaves 7,074 nominal unmasked coordinates.

Representation / composition / objective	n_Θ	Source test	Centre 1	Centre 2
Full EF, paired CE (reference)	10,550	0.9732	0.9356	0.9453
Full EF, direct contractions	10,550	0.9721	0.9368	0.9447
Moment-based Gaussian closure	10,550	0.9778	0.9449	0.9669
No explicit parent correction	10,550	0.9631	0.9436	0.9400
Node-only covering	1,858	0.9666	0.9292	0.9457
Node-only, enlarged capacity	10,558	0.9659	0.9326	0.9559
Full EF, reduced capacity	7,074	0.9620	0.9443	0.9365
Single fixed partition	10,550	0.9622	0.9385	0.9554
Original signed objective (reference)	10,550	0.9608	0.9432	0.9451
Class-centred signed objective	10,550	0.9598	0.9420	0.9494

Native outputs and reproducibility. Each input returns probability/review status, all member/class supports/types, signed anchors, four corrections and dependencies. Table 1 covers 340 images, all 17 held-out patients and three seeds/views. The comparators lack this native covering-program contract; clinical relevance needs separate assessment. All 126 prediction files bind state/input/patient identities; 949 payloads and 336 metrics reproduce within 8.25×10^{-13}. Centred-model 50% nominal source retention transfers to 39.81%/18.14%, with error 0.99%/0.86%; this is not a safety guarantee. Variant-specific aligned stability/full replay, isolated GPU energy and matched baseline tuning remain unmeasured.

10 Conclusion

Evidence Field combines image-relative encoding and exact typed covering for 2D/3D grids. Histopathology results exceed the tested from-scratch baselines at both held-out hospitals, with reconstructible signed evidence. Mechanism benefits are mixed. These 2D measurements and mathematical contracts support further modality-specific and clinical evaluation.

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